Problem Statement 3: Autonomous Web3 Research and On-Chain Settlement Agents

The convergence of Artificial Intelligence and decentralized blockchain infrastructure solves a critical vulnerability in the AI ecosystem: verifiable execution and autonomous resource management. Standard AI agents operating on traditional Web2 infrastructure are constrained by human-in-the-loop payment systems, such as credit card billing, and operate on opaque centralized servers where the integrity of the compute cannot be guaranteed. This problem statement requires the construction of a decentralized network of autonomous AI agents capable of collaborating, negotiating, and transacting value without human intermediaries, ensuring that the outputs of their computations are mathematically verified.
Drawing deep inspiration from projects showcased at ETHGlobal, this system takes the form of an "Intelligence Exchange" or a decentralized federated learning platform. In this environment, human users or other smart contracts post a bounty for a complex research task. AI agents bid on the task, execute the work using Trusted Execution Environments (TEEs) to prove they ran the specific model requested without tampering, and receive automated settlement via smart contracts upon successful completion[22].

Module	Architectural Domain	Technical Deliverables and Specifications	Integration Interfaces
Team A	**Blockchain & Smart Contracts**	Decentralized ledger logic for escrow, identity, and settlement. Solidity smart contracts defining tokenomics (e.g., 81% worker yield, 9% staker yield, 10% treasury distribution)[23]. Integration with decentralized storage networks like 0G Storage or Filecoin for model artifacts[4].	Exposes Web3 RPC endpoints. Emits blockchain events indicating task creation or funding, consumed by Team B.

Team B	**Verifiable AI Execution (TEE/zkML)**	Off-chain AI inference management generating cryptographic proofs of correct execution. Integration with Trusted Execution Environments. Orchestration of federated learning pipelines using frameworks like Flower[4].	Listens to Team A's smart contract events. Submits execution proofs and Content Identifiers (CIDs) back to the blockchain.
Team C	**Agent Logic & Market Interface**	Construction of the autonomous agents that participate in the market. LangChain/AgentKit implementations for wallet management[24]. Next.js Web3 frontend utilizing wagmi and viem for human observers to fund bounties[25].	Interacts with Team A's contracts via cryptographic wallets. Sends execution payloads to Team B's secure enclaves.

The technical friction in Web3 AI is the "Oracle Problem"—the mechanism by which the blockchain verifies the truth of off-chain data and compute. Team B must grapple with TEEs (such as Intel SGX) or zero-knowledge machine learning (zkML) to ensure that the LLM output returned to the smart contract was legitimately generated by the requested model, and not spoofed by a malicious node attempting to claim the bounty with random data[22]. Furthermore, latency is a severe challenge. Blockchain settlement times and decentralized storage retrieval can introduce significant delays into the agent workflow. Team C must design a robust frontend that elegantly handles asynchronous loading states, utilizing subgraphs to index blockchain events for real-time UI updates[3].

To execute this, students can reference Bhavik-punmiya/ethglobal-trifeacta, a decentralized platform where machine learning models are evaluated securely within a TEE. The accuracy

score is computed in a tamper-proof enclave, and the results trigger automated prize payouts via Solidity smart contracts[22]. This provides a masterclass in bridging off-chain compute with on-chain verification. Additionally, ivandda/ethglobal-ai-federated-learning-project demonstrates on-chain verification of federated learning training rounds, utilizing the Flower framework and storing heavy model weights on decentralized storage while recording the CIDs immutably on an EVM-compatible chain[4]. For agent interaction logic, fractaldotbox/civ-ai-ethglobal showcases a simulation where on-chain AI agents collaborate to research Artificial General Intelligence, utilizing real-world data like weather to optimize resource generation and tracking reputations entirely on-chain[26]. Finally, the Coinbase AgentKit allows AI agents to automate fund locking and execute smart contract approvals seamlessly, giving agents their own cryptographic wallets to autonomously pay for APIs or reward developers[24].